

Quantum Kerr OAM

Ann Author* and Second Author†

Authors' institution and/or address

*This line break forced with *

(MUSO Collaboration)

Charlie Author‡

Second institution and/or address

This line break forced and

Third institution, the second for Charlie Author

Delta Author

Authors' institution and/or address

*This line break forced with *

(CLEO Collaboration)

(Dated: June 11, 2026)

An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

I. SPATIALLY STRUCTURED BOSONIC FIELD

Consider a bosonic quantum field $\mathcal{E}(\mathbf{r})$ satisfying the commutation relations

$$[\mathcal{E}(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)] = \delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2), \quad [\mathcal{E}(\mathbf{r}_1), \mathcal{E}(\mathbf{r}_2)] = [\mathcal{E}^\dagger(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)] = 0 \quad (1)$$

In the above equation and in what follows, $\mathbf{r} = (x, y)$ is the transverse coordinate.

One can introduce ladder operators $a_v, a_v^\dagger$ associated with the spatial mode v by the expression

$$a(v) = \int_{\mathbb{R}^2} v^*(\mathbf{r}) \mathcal{E}(\mathbf{r}) d\mathbf{r}. \quad (2)$$

We then have

$$[a(v_1), a(v_2)^\dagger] = \int_{\mathbb{R}^2} v_1^*(\mathbf{r}) v_2(\mathbf{r}) d\mathbf{r} = \langle v_1 | v_2 \rangle, \quad (3)$$

while $[a(v_1), a(v_2)] = [a(v_1)^\dagger, a(v_2)^\dagger] = 0$.

Let us define the quadrature associated to the spatial mode v as

$$\begin{aligned} X(v) &= \frac{1}{\sqrt{2}} \int_{\mathbb{R}^2} v^*(\mathbf{r}) \mathcal{E}(\mathbf{r}) + v(\mathbf{r}) \mathcal{E}^\dagger(\mathbf{r}) d\mathbf{r} \\ &= \frac{a(v) + a(v)^\dagger}{\sqrt{2}} \end{aligned} \quad (4)$$

We then have the commutation relation

$$[X(v_1), X(v_2)] = i \operatorname{Im} \langle v_1 | v_2 \rangle, \quad (5)$$

which leads to the uncertainty principle

$$\Delta X(v_1)^2 \Delta X(v_2)^2 \geq \frac{|\operatorname{Im} \langle v_1 | v_2 \rangle|^2}{4}. \quad (6)$$

* Also at Physics Department, XYZ University.

† Second.Author@institution.edu

‡ <http://www.Second.institution.edu/~Charlie.Author>

One could also introduce the complementary quadrature $P(v) = X(-iv)$ so that

$$[X_{v_1}, P_{v_2}] = i \operatorname{Re} \langle v_1 | v_2 \rangle \quad (7)$$

and

$$\Delta X(v_1)^2 \Delta P(v_2)^2 \geq \frac{|\operatorname{Re} \langle v_1 | v_2 \rangle|^2}{4}. \quad (8)$$

We are interested in observables of the form

$$C(v_1, v_2) = \frac{1}{2} \langle \{X(v_1), X(v_2)\} \rangle - \langle X(v_1) \rangle \langle X(v_2) \rangle = \operatorname{Re} \left[\frac{\langle v_1 | v_2 \rangle}{2} + f(v_1, v_2) + h(v_1, v_2) \right], \quad (9)$$

where we have defined

$$\begin{aligned} f(v_1, v_2) &= \langle a(v_1) a(v_2) \rangle - \langle a(v_1) \rangle \langle a(v_2) \rangle \\ h(v_1, v_2) &= \langle a(v_2)^\dagger a(v_1) \rangle - \langle a(v_2)^\dagger \rangle \langle a(v_1) \rangle \end{aligned} \quad (10)$$

In terms of the fields, the required quantities are written as

$$f(v_1, v_2) = \int v_1^*(\mathbf{r}_2) v_2^*(\mathbf{r}_1) F(\mathbf{r}_1, \mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2; \quad F(\mathbf{r}_1, \mathbf{r}_2) = \langle \mathcal{E}(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \rangle - \langle \mathcal{E}(\mathbf{r}_1) \rangle \langle \mathcal{E}(\mathbf{r}_2) \rangle \quad (11)$$

$$h(v_1, v_2) = \int v_1^*(\mathbf{r}_1) v_2(\mathbf{r}_2) H(\mathbf{r}_1, \mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2; \quad H(\mathbf{r}_1, \mathbf{r}_2) = \langle \mathcal{E}^\dagger(\mathbf{r}_2, z) \mathcal{E}(\mathbf{r}_1, z) \rangle - \langle \mathcal{E}^\dagger(\mathbf{r}_2, z) \rangle \langle \mathcal{E}(\mathbf{r}_1, z) \rangle \quad (12)$$

This formulation is useful because f and g are normally ordered expectation values, which can be directly obtained from stochastic simulations in the positive P representation. Furthermore, this representation also makes it easier to perform linear combinations of the modes, which will be useful later. These combinations will be based on the properties

$$\begin{aligned} f(v_1, v_2 + \lambda v_3) &= f(v_1, v_2) + \lambda^* f(v_1, v_3) \\ f(v_1, v_2) &= f(v_2, v_1) \\ h(v_1, v_2 + \lambda v_3) &= h(v_1, v_2) + \lambda h(v_1, v_3) \\ h(v_1, v_2) &= h(v_2, v_1)^* \end{aligned} \quad (13)$$

which are a reflection of the fact that $a(v)$ is antilinear while $a^\dagger(v)$ is linear.

A. One mode correlation

Let us analyze the correlation matrix $\sigma_{jk} = \frac{1}{2} \langle \{Z_j, Z_k\} \rangle - \langle Z_j \rangle \langle Z_k \rangle$ where $\mathbf{Z} = (X(v), P(v))$. We observe that $h(v, v) \in \mathbb{R}$, so we get

$$\sigma = \begin{pmatrix} \frac{1}{2} + h(v, v) & 0 \\ 0 & \frac{1}{2} + h(v, v) \end{pmatrix} + \begin{pmatrix} \operatorname{Re} f(v, v) & -\operatorname{Im} f(v, v) \\ -\operatorname{Im} f(v, v) & -\operatorname{Re} f(v, v) \end{pmatrix}. \quad (14)$$

Notice that $f(e^{i\phi}v, e^{i\phi}v) = e^{-i2\phi} f(v, v)$. Therefore, we can diagonalize σ by choosing $\phi = [\arg f(v, v) \pm \pi]/2$. We then get eigenvalues

$$\lambda_{\pm} = \frac{1}{2} + h(v, v) \pm |f(v, v)|. \quad (15)$$

We then see that the quadrature rotated along the axis ϕ has minimum variance of λ_- , while the one rotated along the axis $\phi + \pi/2$ has maximum variance λ_+ . Furthermore, we have squeezing if $|f(v, v)| > h(v, v)$.

B. Duan Criterion

Consider two orthonormal states v_1, v_2 , i.e., $\langle v_j | v_k \rangle = \delta_{jk}$. The Duan criterion [1] states that if

$$D = \left\langle \Delta [X(v_1) + X(v_2)]^2 \right\rangle + \left\langle \Delta [P(v_1) - P(v_2)]^2 \right\rangle < 2, \quad (16)$$

then the states are entangled.

Let us define $v_{\pm} = v_1 \pm v_2$. We then have

$$D = \langle \Delta X(v_+)^2 \rangle + \langle \Delta P(v_-)^2 \rangle. \quad (17)$$

Although $v_{\pm}$ is not normalized, (9) still holds. We then obtain

$$\begin{aligned} D &= 2 + h(v_+, v_+) + h(v_-, v_-) + \text{Re}[f(v_+, v_+) - f(v_-, v_-)] \\ &= 2[1 + h(v_1, v_1) + h(v_2, v_2)] + 4\text{Re}f(v_1, v_2) \end{aligned} \quad (18)$$

Once again, we can maximize the possible violation by rotating the quadrature of the modes, say, v_2 by an angle $\phi = [\arg f(v_1, v_2) \pm \pi]/2$. We get an optimal possible violation of

$$D_{\text{opt}} = 2[1 + h(v_1, v_1) + h(v_2, v_2)] - 4|f(v_1, v_2)| \quad (19)$$

II. THE KERR SYSTEM

A. Heisenberg Equation for the Field and Linear Solution

We will assume that the field is propagating along the z direction, and satisfies the Heisenberg equation of motion

$$2ik\partial_z \mathcal{E}(\mathbf{r}, z) + \nabla^2 \mathcal{E}(\mathbf{r}, z) = g\mathcal{E}^\dagger(\mathbf{r}, z)\mathcal{E}^2(\mathbf{r}, z). \quad (20)$$

Let us decompose $\mathcal{E} = \mathcal{E}_0 + \delta\mathcal{E}$ where $\mathcal{E}_0$ is the solution of

$$2ik\partial_z \mathcal{E}_0(\mathbf{r}, z) + \nabla^2 \mathcal{E}_0(\mathbf{r}, z) = 0, \quad \mathcal{E}_0(\mathbf{r}, 0) = \mathcal{E}(\mathbf{r}, 0). \quad (21)$$

Let u be a solution of the paraxial wave equation $2ik\partial_z u(\mathbf{r}, z) + \nabla^2 u(\mathbf{r}, z) = 0$. Then, the annihilation operator corresponding to mode u is given by

$$a(u, z) = \int u^*(\mathbf{r}, z)\mathcal{E}(\mathbf{r}, z)d\mathbf{r} = a_0(u) + \delta a(u, z) \quad (22)$$

where

$$a_0(u) = \int u^*(\mathbf{r}, z)\mathcal{E}_0(\mathbf{r}, z)d\mathbf{r} \quad (23)$$

is independent of z (take a z derivative, use the equations of motion and integrate by parts) and

$$\delta a(u, z) = \int u^*(\mathbf{r}, z)\delta\mathcal{E}(\mathbf{r}, z)d\mathbf{r} \quad (24)$$

Our initial state will be a coherent one:

$$|\alpha\rangle = D(\alpha)|0\rangle = \exp(a_0(\alpha)^\dagger - \alpha^* a(\alpha))|0\rangle. \quad (25)$$

Here, $\alpha(\mathbf{r})$ is a non normalized spatial function, with norm $\|\alpha\|$. This state is an eigenvector of the free field:

$$\mathcal{E}_0(\mathbf{r}, z)|\alpha\rangle = \alpha(\mathbf{r}, z)|\alpha, u\rangle \quad (26)$$

The variation $\delta\mathcal{E}$ satisfies

$$2ik\partial_z \delta\mathcal{E}(\mathbf{r}, z) + \nabla^2 \delta\mathcal{E}(\mathbf{r}, z) = g\mathcal{E}^\dagger(\mathbf{r}, z)\mathcal{E}^2(\mathbf{r}, z). \quad (27)$$

Evaluating this at $z = 0$ gives us

$$2ik\partial_z\delta\mathcal{E}(\mathbf{r}, 0) = g\mathcal{E}_0^\dagger(\mathbf{r}, 0)\mathcal{E}_0^2(\mathbf{r}, 0) \quad (28)$$

from which we obtain the Taylor expansion

$$\delta\mathcal{E}(\mathbf{r}, z) \approx -\frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r}, 0)\mathcal{E}_0^2(\mathbf{r}, 0) \quad (29)$$

For the complete field we then have

$$\mathcal{E}(\mathbf{r}, z) \approx \mathcal{E}_0(\mathbf{r}, 0) - \frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r}, 0)\mathcal{E}_0^2(\mathbf{r}, 0) \quad (30)$$

This expression will be the basis of our further developments.

B. Expectation value of field

Taking expectation values, we arrive at

$$\langle\mathcal{E}(\mathbf{r}, z)\rangle \approx \alpha(\mathbf{r}, 0) - \frac{igz}{2k}|\alpha(\mathbf{r}, 0)|^2\alpha(\mathbf{r}, 0). \quad (31)$$

The expectation value of the annihilation operator a_v corresponding to a spatial mode v is then

$$\langle a_v \rangle = \langle v|\alpha \rangle - \frac{igz}{2k}\langle v, \alpha|\alpha, \alpha \rangle \quad (32)$$

In the above expression we have introduced the second spatial overlap

$$\langle v_1, v_2|u_1, u_2 \rangle = \int v_1^*(\mathbf{r}, z)v_2^*(\mathbf{r}, z)u_1(\mathbf{r}, z)u_2(\mathbf{r}, z)d\mathbf{r}. \quad (33)$$

When $\alpha = \|\alpha\|u_{0l}$ is a Laguerre-Gaussian mode with radial order $p = 0$ and topological charge l , we have that

$$|u_{0l}(\mathbf{r}, z)|u_{0l}^2(\mathbf{r}, z) = \sum_{q=0}^{|l|} C_{ql}\bar{u}_{ql}(\mathbf{r}, z) \quad (34)$$

where $\bar{u}_{ql}$ are Laguerre-Gaussian modes with a reduced waist. Therefore, one may write

$$\langle a_v \rangle = \|\alpha\| \langle v|u_{0,l} \rangle - \frac{igz\|\alpha\|^3}{2k} \sum_{q=0}^{|l|} C_{ql} \langle v|\bar{u}_{ql} \rangle \quad (35)$$

In particular, when $v = \bar{u}_{ql}$, we have that

$$\langle a_{\bar{u}_{ql}} \rangle = \begin{cases} \|\alpha\| \langle \bar{u}_{ql}|u_{0,l} \rangle - \frac{igz\|\alpha\|^3 C_{ql}}{2k} & q \leq |l| \\ \|\alpha\| \langle \bar{u}_{ql}|u_{0,l} \rangle & q > |l| \end{cases}. \quad (36)$$

C. Correlation functions

One of the two point field function is

$$\mathcal{E}^\dagger(\mathbf{r}_1, z)\mathcal{E}(\mathbf{r}_2, z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z)\delta\mathcal{E}(\mathbf{r}_2, z) + \delta\mathcal{E}^\dagger(\mathbf{r}_1, z)\mathcal{E}_0(\mathbf{r}_2, z) \quad (37)$$

where we have only kept terms linear in $\delta\mathcal{E}$.

Close to $z = 0$, we have that

$$\mathcal{E}^\dagger(\mathbf{r}_1, z)\mathcal{E}(\mathbf{r}_2, z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1, 0)\mathcal{E}_0(\mathbf{r}_2, 0) - \frac{igz}{2k} \left[\mathcal{E}_0^\dagger(\mathbf{r}_1, 0)\mathcal{E}_0^\dagger(\mathbf{r}_2, 0)\mathcal{E}_0^2(\mathbf{r}_2, 0) - \mathcal{E}_0^\dagger(\mathbf{r}_1, 0)^2\mathcal{E}_0(\mathbf{r}_1, 0)\mathcal{E}_0(\mathbf{r}_2, 0) \right] \quad (38)$$

Evaluating this at our initial state gives us

$$\begin{aligned}\langle \mathcal{E}^\dagger(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \rangle &\approx \alpha^*(\mathbf{r}_1, 0) \alpha(\mathbf{r}_2, 0) \left\{ 1 - \frac{igz}{2k} \left[|\alpha(\mathbf{r}_2, 0)|^2 - |\alpha(\mathbf{r}_1, 0)|^2 \right] \right\} \\ &\approx \langle \mathcal{E}^\dagger(\mathbf{r}_1, z) \rangle \langle \mathcal{E}(\mathbf{r}_2, z) \rangle\end{aligned}\quad (39)$$

where the last approximate inequality comes from direct comparison with (31). In particular, this shows that, for any v_1 and v_2 , we have $H(\mathbf{r}_1, \mathbf{r}_2) \approx 0$.

We can also use this to write the expectation value of the number operator:

$$\langle n(v) \rangle = \langle a^\dagger(v) a(v) \rangle \approx |\langle a(v) \rangle|^2 \approx |\langle v | \alpha \rangle|^2 + \frac{gz}{k} \text{Im} \langle v | \alpha \rangle \langle \alpha, \alpha | v \rangle \quad (40)$$

When both α and v are arbitrary Laguerre-Gaussian modes (any p, l, w), we have that $\text{Im} \langle \alpha | u \rangle \langle \alpha, \alpha | v \rangle = 0$, so there is no first order correction to the occupation number.

The other two point field function is

$$\begin{aligned}\mathcal{E}(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) &\approx \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) + \mathcal{E}_0(\mathbf{r}_1, z) \delta \mathcal{E}(\mathbf{r}_2, z) + \delta \mathcal{E}(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) \\ &\approx \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) - \frac{igz}{2k} \left[\mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0^\dagger(\mathbf{r}_2, z) \mathcal{E}_0^2(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z) \mathcal{E}_0^2(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) \right] \\ &= \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) - \frac{igz}{2k} \left[\mathcal{E}_0^\dagger(\mathbf{r}_2, z) \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0^2(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z) \mathcal{E}_0^2(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) + \delta(\mathbf{r}_1 - \mathbf{r}_2) \mathcal{E}_0^2(\mathbf{r}_2, z) \right]\end{aligned}\quad (41)$$

The appearance of the delta function here is important, because now

$$\begin{aligned}\langle \mathcal{E}(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \rangle &\approx \alpha(\mathbf{r}_1, z) \alpha(\mathbf{r}_2, z) \left[1 - \frac{igz}{2k} \left[|\alpha(\mathbf{r}_2, z)|^2 + |\alpha(\mathbf{r}_1, z)|^2 \right] \right] - \frac{igz}{2k} \delta(\mathbf{r}_1 - \mathbf{r}_2) \alpha^2(\mathbf{r}_2, z) \\ &\approx \langle \mathcal{E}(\mathbf{r}_1, z) \rangle \langle \mathcal{E}(\mathbf{r}_2, z) \rangle - \frac{igz}{2k} \delta(\mathbf{r}_1 - \mathbf{r}_2) \alpha^2(\mathbf{r}_2, z)\end{aligned}\quad (42)$$

Therefore, we may conclude that

$$F(\mathbf{r}_1, \mathbf{r}_2) \approx -\frac{igz}{2k} \delta(\mathbf{r}_1 - \mathbf{r}_2) \alpha^2(\mathbf{r}_2, z) \quad (43)$$

and

$$f(v_1, v_2) \approx -\frac{igz}{2k} \langle v_1, v_2 | \alpha, \alpha \rangle. \quad (44)$$

This leads us to a minimum quadrature variance for mode v of

$$\lambda_- = \frac{1}{2} - \frac{z}{2k} |g \langle v, v | \alpha, \alpha \rangle| \quad (45)$$

and an optimal Duan parameter for modes v_1 and v_2 of

$$D_{\text{opt}} = 2 - \frac{2z}{k} |g \langle v_1, v_2 | \alpha, \alpha \rangle|. \quad (46)$$

The corresponding angles are $\phi_{\text{squeezing}} = (-\pi/2 \pm \pi + \arg g \langle v, v | \alpha, \alpha \rangle)/2$ and $\phi_{\text{duan}} = (-\pi/2 \pm \pi + \arg g \langle v_1, v_2 | \alpha, \alpha \rangle)/2$

At this point, it becomes clear that the overlaps of the form $\langle v_1, v_2 | \alpha, \alpha \rangle$ are the ones that dominate this sort of process. On the other hand, the generation of radial orders comes from the properties of the different overlap $\langle v, \alpha | \alpha, \alpha \rangle$, which appears in (32). It is therefore hard to see any special implication of the selection rules here.

Consider the case where all the involved modes are vortices: $u(\mathbf{r}) = u_r(r) e^{il_u \phi}$ and similarly for the v functions. We have squeezing if $l_v = l_u$ and we have entanglement if $l_{v_1} + l_{v_2} = 2l_u$. If all the radial functions are positive, as is the case for $LG_{0,0}$ and $LG_{0,\pm 1}$, and α is positive, then the optimal angles are both $\text{sign}(g)\pi/4$.

III. NUMERICAL CONSIDERATIONS

Note that, according to (1), the field $\mathcal{E}(\mathbf{r})$ has dimensions of $1/\sqrt{\text{area}}$. Therefore, by dimensional analysis on eq. (20), we have that g is dimensionless.

In order to put the equation in a more convenient form, we introduce the dimensionless coordinate $\tilde{\mathbf{r}} = \mathbf{r}/w_0$ where w_0 is the waist of the input beam. We also introduce the dimensionless propagation distance $\tilde{z} = z/z_R$ where $z_R = kw_0^2/2$ is the Rayleigh range. Finally, we introduce the rescaled field $\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) = \sqrt{\Delta A} \mathcal{E}(\mathbf{r}, z)$, where ΔA is the area of a pixel in the numerical grid. In terms of these variables, the commutation relations read

$$[\tilde{\mathcal{E}}(\tilde{\mathbf{r}}), \tilde{\mathcal{E}}(\tilde{\mathbf{r}})^\dagger] \approx \delta_{\tilde{\mathbf{r}}, \tilde{\mathbf{r}}'}, \quad (47)$$

while the equation of motion reads

$$i\partial_{\tilde{z}}\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) = -\frac{1}{4}\tilde{\nabla}^2\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) + \frac{g}{4\Delta\tilde{A}}\tilde{\mathcal{E}}^\dagger(\tilde{\mathbf{r}}, \tilde{z})\tilde{\mathcal{E}}^2(\tilde{\mathbf{r}}, \tilde{z}) \quad (48)$$

where $\Delta\tilde{A} = \Delta A/w_0^2$ is the area of a pixel in the rescaled coordinates.

In the positive P representation, the field $\tilde{\mathcal{E}}$ is represented by two independent complex stochastic fields ψ_1 and ψ_2 satisfying the stochastic equations (compare with [2])

$$\begin{aligned} i\partial_{\tilde{z}}\psi_1(\tilde{\mathbf{r}}, \tilde{z}) &= -\frac{1}{4}\tilde{\nabla}^2\psi_1(\tilde{\mathbf{r}}, \tilde{z}) + \frac{g}{4\Delta\tilde{A}}\psi_2(\tilde{\mathbf{r}}, \tilde{z})\psi_1^2(\tilde{\mathbf{r}}, \tilde{z}) + \sqrt{\frac{ig}{4\Delta\tilde{A}}}\psi_1(\tilde{\mathbf{r}}, \tilde{z})\xi_1(\tilde{\mathbf{r}}, \tilde{z}) \\ -i\partial_{\tilde{z}}\psi_2(\tilde{\mathbf{r}}, \tilde{z}) &= -\frac{1}{4}\tilde{\nabla}^2\psi_2(\tilde{\mathbf{r}}, \tilde{z}) + \frac{g}{4\Delta\tilde{A}}\psi_1(\tilde{\mathbf{r}}, \tilde{z})\psi_2^2(\tilde{\mathbf{r}}, \tilde{z}) + \sqrt{-\frac{ig}{4\Delta\tilde{A}}}\psi_2(\tilde{\mathbf{r}}, \tilde{z})\xi_2(\tilde{\mathbf{r}}, \tilde{z}) \end{aligned} \quad (49)$$

where ξ_1 and ξ_2 are independent real Gaussian white noises with zero mean and correlations

$$\langle \xi_j(\tilde{\mathbf{r}}, \tilde{z}) \xi_k(\tilde{\mathbf{r}}', \tilde{z}') \rangle = \delta_{jk} \delta_{\tilde{\mathbf{r}}, \tilde{\mathbf{r}}'} \delta_{\tilde{z}, \tilde{z}'} \quad (50)$$

For a coherent initial state in the mode u , we have that $\psi_1(\tilde{\mathbf{r}}, 0) = \alpha(\tilde{\mathbf{r}}, 0)$ and $\psi_2(\tilde{\mathbf{r}}, 0) = \alpha^*(\tilde{\mathbf{r}}, 0)$.

IV. PURELY NONLINEAR APPROACH

Let us make a modification to our equations of motion where

$$2ik\partial_z\mathcal{E}(\mathbf{r}, z) = \int_{\mathbb{R}^2} g(\mathbf{s})\mathcal{E}^\dagger(\mathbf{r}-\mathbf{s}, z)\mathcal{E}(\mathbf{r}-\mathbf{s}, z)d\mathbf{s} \mathcal{E}(\mathbf{r}, z). \quad (51)$$

In this form, the equation has a constant of motion $\mathcal{E}^\dagger(\mathbf{r}, z)\mathcal{E}(\mathbf{r}, z)$, which leads us to the exact solution

$$\begin{aligned} \mathcal{E}(\mathbf{r}, z) &= \exp\left[-\frac{iz}{2k}\int_{\mathbb{R}^2} g(\mathbf{s})\mathcal{E}^\dagger(\mathbf{r}-\mathbf{s}, 0)\mathcal{E}(\mathbf{r}-\mathbf{s}, 0)d\mathbf{s}\right] \mathcal{E}(\mathbf{r}, 0) \\ &=: \exp\left[\int_{\mathbb{R}^2} \left(e^{-izg(\mathbf{s})/2k} - 1\right) \mathcal{E}^\dagger(\mathbf{r}-\mathbf{s}, 0)\mathcal{E}(\mathbf{r}-\mathbf{s}, 0)d\mathbf{s}\right] : \mathcal{E}(\mathbf{r}, 0) \end{aligned} \quad (52)$$

In the last line, we used an important result known as the normal ordering theorem [3, 4]. This allows us to simply calculate the expectation value for the field in an initial coherent state:

$$\langle \mathcal{E}(\mathbf{r}, z) \rangle = \exp\left[\int_{\mathbb{R}^2} \left(e^{-izg(\mathbf{s})/2k} - 1\right) |\alpha(\mathbf{r}-\mathbf{s}, 0)|^2 d\mathbf{s}\right] u(\mathbf{r}, 0) \quad (53)$$

We see that (31) is then a first order approximation of (53) with $g(\mathbf{s}) = g_0\delta(\mathbf{s})$. Nonetheless, we see that we get singularities in the full expression when using this form of g . Let us then choose g as proportional to an approximation of the delta function. For definiteness, let us define

$$\delta_0(x, a) = \frac{\Theta(a/2 - |x|)}{a}, \quad (54)$$

which approximates the delta distribution as $a \rightarrow 0$. Here, $\Theta(x)$ is the step function. Let us denote $\delta_0(\mathbf{r}, a) = \delta_0(x, a)\delta_0(y, a)$. We then choose $g(\mathbf{s}) = g_0\delta_0(\mathbf{s})$.

We will usually make the approximation that the support of the response function g is much smaller than the characteristic distances of the mode α . We can then make the approximation

$$\langle \mathcal{E}(\mathbf{r}, z) \rangle = \exp \left[G(z) |\alpha(\mathbf{r}, 0)|^2 \right] \alpha(\mathbf{r}, 0) \quad (55)$$

where

$$G(z) = \int_{\mathbb{R}^2} \left(e^{-izg(\mathbf{s})/2k} - 1 \right) d\mathbf{s} = a^2 \left(e^{-izg_0/2ka^2} - 1 \right) \quad (56)$$

Notice then that, in the general case, this is quite dependent on the value of a . Nonetheless, when $z|g_0|/2ka^2 \ll 1$, we get

$$G(z) \approx -\frac{izg_0}{2k}, \quad (57)$$

which is independent of a .

Following the result of [4] we can calculate

$$\begin{aligned} F(\mathbf{r}_1, \mathbf{r}_2) &\approx \alpha(\mathbf{r}_1, 0)\alpha(\mathbf{r}_2, 0) \exp \left[G(z) (|\alpha(\mathbf{r}_1)|^2 + |\alpha(\mathbf{r}_2)|^2) \right] \\ &\times \left[e^{-ig(\mathbf{r}_1 - \mathbf{r}_2)z/2k + |\alpha(\mathbf{r}_1)|^2 G_1(\mathbf{r}_1 - \mathbf{r}_2)} - 1 \right] \end{aligned} \quad (58)$$

where

$$\begin{aligned} G_1(\mathbf{r}, z) &= \frac{1}{2} \int_{\mathbb{R}^2} (e^{-izg(\mathbf{s} - \mathbf{r})} - 1)(e^{-izg(\mathbf{s})} - 1) d\mathbf{s} \\ &= \frac{a^4 (e^{-ig_0z/2ka^2} - 1)^2}{2} \delta_1(\mathbf{r}, a) \\ &\approx -\frac{(g_0z/2k)^2}{2} \delta_1(\mathbf{r}, a) \end{aligned} \quad (59)$$

where we have defined

$$\delta_1(x, a) = \frac{(a - |x|)\Theta(a - |x|)}{a^2} \quad (60)$$

as another approximation of the delta function.

Notice, then, that, for modes with variation on a scale much larger than a , the second line of (58) is essentially proportional to $\delta(\mathbf{r}_1 - \mathbf{r}_2)$, and we may write

$$F(\mathbf{r}_1, \mathbf{r}_2) \approx \alpha(\mathbf{r}_1, z)^2 \bar{F}(\mathbf{r}_1, a) \delta(\mathbf{r}_1 - \mathbf{r}_2) \quad (61)$$

where

$$\alpha(\mathbf{r}, z) = \alpha(\mathbf{r}, 0) e^{-ig_0z|\alpha(\mathbf{r}, 0)|^2/2k} \quad (62)$$

is the evolved field under the classical Kerr interaction and

$$\bar{F}(\mathbf{r}, a) = \int_{\mathbb{R}^2} \left\{ \exp \left[-i\frac{g_0z}{2k} \delta_0(\mathbf{s}, a) - \frac{1}{2} \left(\frac{g_0z}{2k} \right)^2 \delta_1(\mathbf{s}, a) |\alpha(\mathbf{r})|^2 \right] - 1 \right\} d\mathbf{s} \quad (63)$$

In general, this function presents a strong dependency on a . Nonetheless, if we also assume that $z^2|g_0|^2|\alpha(\mathbf{r})|^2/4k^2a^2 \ll 1$, then we may expand the exponential in a Taylor series and we get

$$\bar{F}(\mathbf{r}, a) = -\frac{g_0z}{2k} \left[i + \frac{g_0|\alpha(\mathbf{r})|^2z}{2k} \right], \quad (64)$$

which does not depend on a .

-
- [1] L.-M. Duan, G. Giedke, J. I. Cirac, and P. Zoller, Inseparability criterion for continuous variable systems, Physical review letters **84**, 2722 (2000).
 - [2] P. Deuar and P. D. Drummond, Correlations in a bec collision: First-principles quantum dynamics with 150 000 atoms, Phys. Rev. Lett. **98**, 120402 (2007).
 - [3] K. J. Blow, R. Loudon, S. J. D. Phoenix, and T. J. Shepherd, Continuum fields in quantum optics, Phys. Rev. A **42**, 4102 (1990).
 - [4] K. J. Blow, R. Loudon, and S. J. D. Phoenix, Exact solution for quantum self-phase modulation, J. Opt. Soc. Am. B **8**, 1750 (1991).